Assignment 1e v2

Future Modeling, Recursive Competency Mapping, and Adaptive Planning

Harvey Chowdhury | MTEC 4502 | Dr. David B. Smith | November 11, 2026

REVIEW DRAFT - proposed assessments and targets await review. Zotero saving is pending.

1. Future-of-work synthesis

My goal is to become an applied AI engineer who builds agentic systems, retrieval applications, and automation tools. I also want to connect that work with game design, visual design, sound, and video. Game design was my original ambition. My portfolio now includes a small browser game, gesture-controlled experiences, an audio story, and workout applications. Those projects give me a starting point, but they do not establish that I am ready for every engineering role.

In Assignment 1d, I clarified that automation and integration matter more to me than choosing a polished job title. I still agree with that direction. What has changed is the evidence available to explain it. My contribution to Tesseract gives me an example of working on an existing shared-memory system. JimBro and my creative projects show other ways I connect an idea with an interactive result. I need a strategy that develops the engineering underneath those experiences.

The World Economic Forum expects AI-related skills to grow in importance while analytical thinking, creativity, and adaptability remain valuable [S1]. Its forecast reaches 2030. For the following five to ten years, approximately 2031 to 2036, I treat changes as scenarios rather than predictions. One possibility is that applied engineers spend less time assembling standard components and more time defining problems, evaluating results, and maintaining systems. Another is that roles split between general integration work and deeper specializations in retrieval, evaluation, or interactive tools.

Anthropic's coding study shows substantial delegation to coding agents, including workflows that still involve human feedback [S2]. That supports my interest in directing systems, but the study does not prove that generated code is correct or that developers become more productive. Repetitive scaffolding, simple interface assembly, and routine asset variations may become less distinctive portfolio evidence. Debugging, data handling, evaluation, and explaining why a system behaves as it does remain priorities in my plan.

My interest in games needs the same scrutiny. GDC's survey reports both generative-AI use and strong negative sentiment, alongside employment concerns [S3]. Easier prototyping does not guarantee a good player experience or a job. I want to build beyond Wasp Dodge+, but I need to demonstrate mechanics, feedback, iteration, and testing. My creative background can help me identify useful applications, rather than assume every game task benefits from automation.

My most robust strategy is connecting technical systems with practical and creative problems. Tesseract began with a need I could explain: several agents should share useful memory without repeated interruptions. I tested the original setup, investigated weaknesses, reviewed code, and directed changes. My merged contribution addresses concurrent note creation and agent identity [E1]. In my own setup, I report that daily maintenance, live writing, and reduced interruptions work. Those reports are different from a controlled benchmark or proof of other users' results.

The vulnerable part is retrieval quality. I sometimes received outdated or less relevant memory, and I have not implemented a specific fix. My planned Rust system will investigate dated records, conflicting facts, and verified updates. Newer records cannot automatically count as more accurate. I need tests that distinguish when information was recorded from when it became valid, and retain uncertainty when a conflict cannot be resolved.

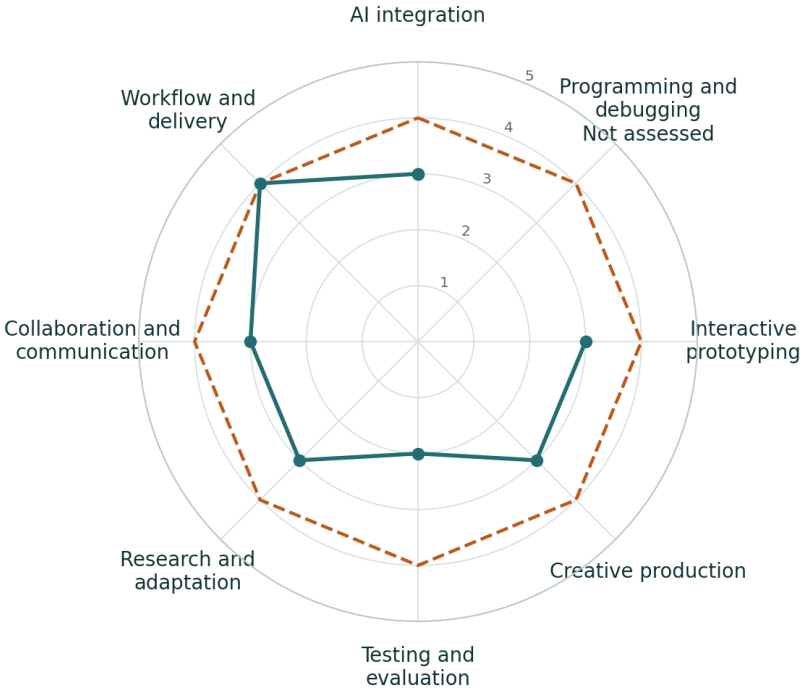
My revised chart therefore separates integration from evaluation and leaves unverified abilities unassessed. An applied-AI role example also expects programming, retrieval, reliable prototypes, and communication [S4]. I will first build a small memory-evaluation set, then a bounded Rust retrieval prototype. A later game prototype can test the same engineering habits in a creative setting. I will review the chart after each milestone instead of raising scores simply because I have added another feature.

Synthesis: 599 words. References and tables are excluded.

2. Role-level competency map

These are provisional assessments for review, not employer scores or grades. They describe the evidence available on November 11, 2026. Tools and collaborators are permitted within a competency when their contribution is explained. Project code alone does not establish individual programming depth. Not assessed means evidence is insufficient, not that ability is zero. Targets are proposed development goals by May 2028 and are not employer requirements. A parent score is a holistic judgment, not the arithmetic mean of its sub-radar. The lower evaluation score reflects informal testing evidence rather than the sophistication of the software.

Applied AI engineering and creative applications



Gaps in the current line mean Not assessed. Missing evidence is never plotted as zero.

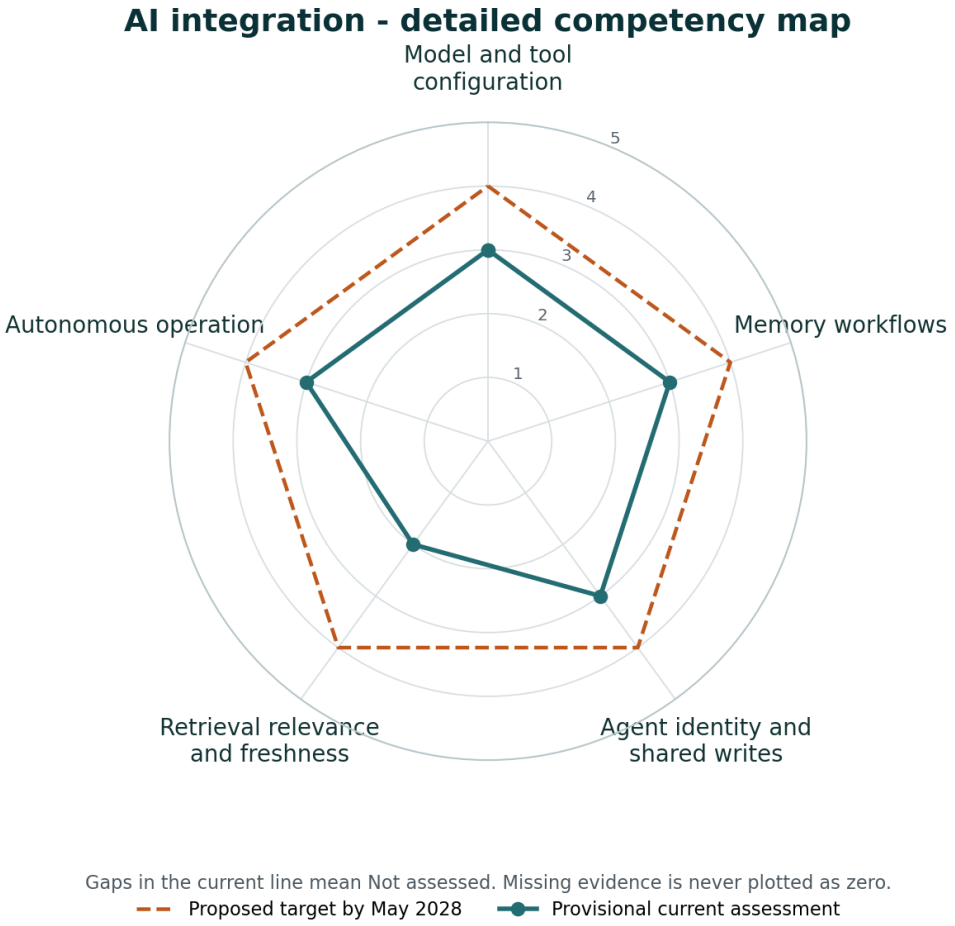
Proposed target by May 2028 — Provisional current assessment

Proposed development goals by May 2028. Not assessed values are omitted from the current line.

Axis	Current	Target	Evidence
AI integration	3	4	E1, E7, E8 and student setup report
Programming and debugging	Not assessed	4	E1, E3, E6 and student code-review account
Interactive prototyping	3	4	E3, E4, E6, E9, E10
Creative production	3	4	E9, E10, E11, E12
Testing and evaluation	2	4	Student Tesseract stress tests, E1 and E3 test artifacts
Research and adaptation	3	4	Prior 1b job analysis, 1d decisions and Tesseract investigation
Collaboration and communication	3	4	E1, E2, E3 reflection and E12
Workflow and delivery	4	4	E3, E5-E12 and repository release records

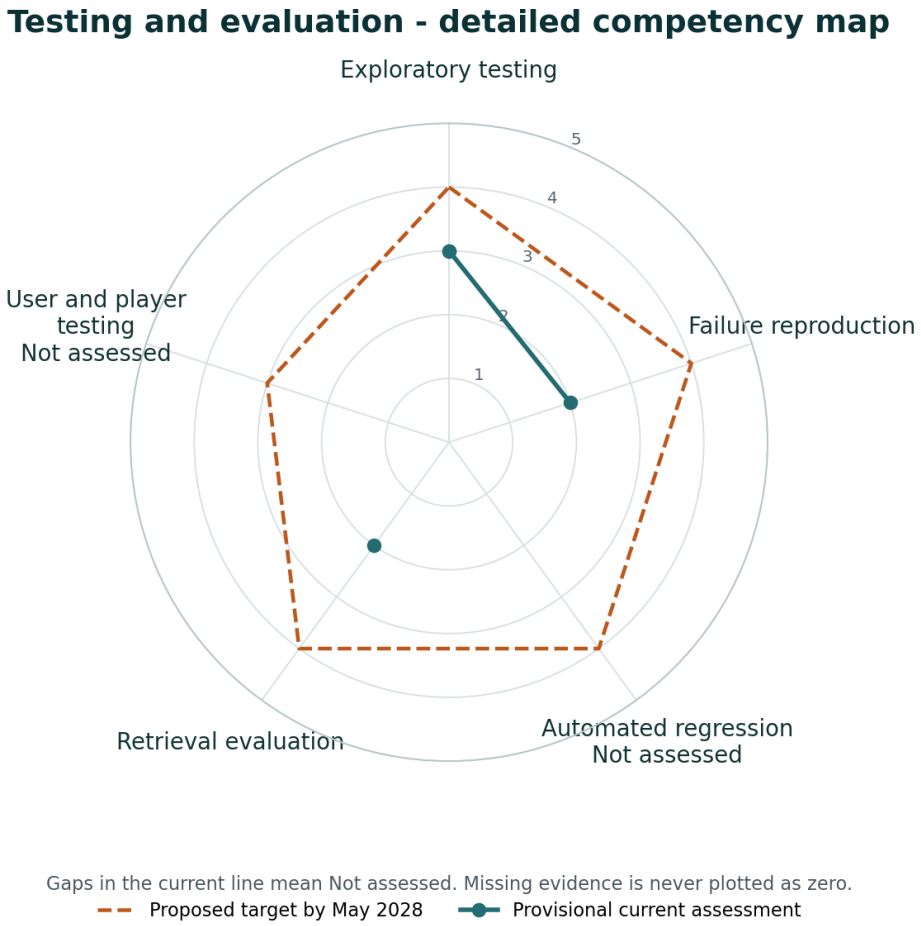
3. Recursive competency maps

AI integration



Provisional current assessments and proposed May 2028 targets. Missing current evidence is shown as a gap. See the matching rubrics for specific behavior and evidence.

Testing and evaluation



Provisional current assessments and proposed May 2028 targets. Missing current evidence is shown as a gap. See the matching rubrics for specific behavior and evidence.

4. Competency rubrics

A level describes demonstrated performance. Tools and collaborators can contribute when their roles are explained. The parent assessment is holistic, not a calculated average.

M1 - AI integration

Observable behavior	Connect existing models, memory, and tools into a usable application.
1	Cannot yet connect a model or tool in a guided exercise.
2	Connects one component with guidance but cannot trace the full flow.
3	Demonstrates a bounded integration and explains its input and output.
4	Reproduces integrations consistently and handles dependency failures.
5	Leads reusable integration architecture and validates adoption by others.
Current / target	3 / 4
Evidence	E1, E7, E8 and student setup report
Rationale	A merged multi-agent contribution and creative model integrations support basic integration. Wider reliability remains unverified.
Next action	Document one complete integration with repeatable setup and failure tests.

M2 - Programming and debugging

Observable behavior	Trace behavior, isolate a defect, and justify a code change.
1	Cannot yet trace a simple program or locate an identified defect.
2	Makes guided edits and recognizes basic error messages.
3	Explains a module and fixes a bounded defect with a passing test.
4	Diagnoses unfamiliar defects and checks regressions across modules.
5	Leads difficult debugging and teaches the reasoning to collaborators.
Current / target	Not assessed / 4
Evidence	E1, E3, E6 and student code-review account
Rationale	Repository code exists and code inspection is reported. A direct walkthrough is still needed to assess individual programming depth.
Next action	Walk through one defect and fix with a reproducible test and explanation.

Competency rubrics - continued

M3 - Interactive prototyping

Observable behavior	Build a task or play loop with clear input, state, and feedback.
1	Cannot yet assemble an interactive sequence.
2	Creates isolated controls with incomplete state transitions.
3	Produces a working bounded flow with understandable feedback.
4	Refines complete flows through observed use and edge-case testing.
5	Leads reusable interaction patterns validated across users and projects.
Current / target	3 / 4
Evidence	E3, E4, E6, E9, E10
Rationale	Several working project structures support basic competence. Recent human validation is not established by this inspection.
Next action	Observe another person completing a task and revise one confusing interaction.

M4 - Creative production

Observable behavior	Shape visual or audio material to communicate an intended experience.
1	Cannot yet prepare a usable visual or audio asset.
2	Produces simple assets with inconsistent technical or narrative control.
3	Creates a coherent bounded piece and documents production decisions.
4	Iterates consistent media quality against audience and delivery needs.
5	Directs a cohesive production and mentors others through its pipeline.
Current / target	3 / 4
Evidence	E9, E10, E11, E12
Rationale	Published visuals and a documented audio story support a bounded production claim. The music video remains uninspected.
Next action	Publish a before-and-after edit with rationale and audience feedback.

Competency rubrics - continued

M5 - Testing and evaluation

Observable behavior	Define success, challenge failures, and compare observed outcomes.
1	Cannot yet identify a checkable success or failure condition.
2	Explores failures and records observations informally.
3	Runs repeatable cases with expected results and records fixes.
4	Combines regression, retrieval, and usability evidence across releases.
5	Designs evaluation strategy and independently audits coverage and limits.
Current / target	2 / 4
Evidence	Student Tesseract stress tests, E1 and E3 test artifacts
Rationale	Exploratory stress testing is described. Existing automated test artifacts do not establish which evaluations the student personally ran.
Next action	Convert exploratory checks into a repeatable test set with expected outcomes.

M6 - Research and adaptation

Observable behavior	Compare evidence and use it to justify a technical or career decision.
1	Cannot yet locate or identify a relevant source.
2	Collects sources but accepts claims without checking fit.
3	Uses relevant sources and explains a decision with stated limitations.
4	Triangulates conflicting sources and tests uncertain technical claims.
5	Builds reusable research methods and helps others evaluate evidence.
Current / target	3 / 4
Evidence	Prior 1b job analysis, 1d decisions and Tesseract investigation
Rationale	Prior career analysis and the reported investigation show evidence-led iteration. Current source selection still needs final student review.
Next action	Compare two retrieval approaches with citations, tradeoffs, and a small experiment.

Competency rubrics - continued

M7 - Collaboration and communication

Observable behavior	Explain contributions, coordinate changes, and respond to feedback.
1	Cannot yet explain role or communicate a proposed change.
2	Shares progress but contribution boundaries remain unclear.
3	Explains a bounded contribution and responds to review or feedback.
4	Coordinates multi-person changes with clear handoffs and documented outcomes.
5	Leads collaboration and develops practices adopted by others.
Current / target	3 / 4
Evidence	E1, E2, E3 reflection and E12
Rationale	The merged pull request and clarified contributor role support a bounded collaboration claim, not team leadership.
Next action	Add a concise contribution case study linking request, review, change, and outcome.

M8 - Workflow and delivery

Observable behavior	Package, publish, document, and maintain usable project versions.
1	Cannot yet produce a runnable or shareable project version.
2	Shares a project that depends on undocumented setup.
3	Publishes a usable version with setup and limitation notes.
4	Repeatedly delivers projects with organized source and release evidence.
5	Leads a reproducible delivery process that others can maintain independently.
Current / target	4 / 4
Evidence	E3, E5-E12 and repository release records
Rationale	Multiple live projects and dated release records support a provisional four. Maintenance by other users has not been demonstrated.
Next action	Test one clean installation or recovery with another user.

Competency rubrics - continued

A1 - Model and tool configuration

Observable behavior	Configure a model or service and explain its operational inputs.
1	Cannot yet configure a guided model call.
2	Runs a sample configuration without explaining its options.
3	Configures a bounded service and explains key options and outputs.
4	Reproduces configuration and manages version or dependency changes.
5	Creates reusable configuration standards validated by other users.
Current / target	3 / 4
Evidence	E7, E8 and student Tesseract setup
Rationale	Creative projects integrate existing vision models. This is application evidence, not evidence of model training.
Next action	Reproduce configuration from a clean documented setup.

A2 - Memory workflows

Observable behavior	Describe and implement when agents read, write, and maintain memory.
1	Cannot yet trace where a memory record is stored.
2	Reads or writes a sample note with guidance.
3	Demonstrates a bounded write, retrieval, and maintenance flow.
4	Checks persistence and recovery across repeated sessions.
5	Designs audited memory lifecycles used reliably by other teams.
Current / target	3 / 4
Evidence	E1 and student report of daily maintenance and live writes
Rationale	The student reports a working personal setup. No independent daily-scheduler run was observed in this review.
Next action	Record a repeatable write-to-recall journey and recovery case.

Competency rubrics - continued

A3 - Agent identity and shared writes

Observable behavior	Preserve contributor identity and prevent conflicting note creation.
1	Cannot yet distinguish contributors or detect a collision.
2	Labels agents manually without checking concurrent writes.
3	Implements a bounded identity or collision fix with test evidence.
4	Reproduces concurrent behavior across supported platforms and failure paths.
5	Designs shared-write contracts and validates independent multi-agent deployments.
Current / target	3 / 4
Evidence	E1 and its multi-agent test file
Rationale	Merged PR #21 specifically addresses identity and create collisions. Tests were inspected, not rerun here.
Next action	Reproduce concurrent writes and inspect identity metadata.

A4 - Retrieval relevance and freshness

Observable behavior	Select relevant current evidence while retaining historical context.
1	Cannot yet explain whether a retrieved item answers a question.
2	Recognizes irrelevant or stale recall but lacks a measured correction.
3	Evaluates a repeatable query set and implements one verified improvement.
4	Handles current, historical, and conflicting evidence with measured results.
5	Leads temporal retrieval evaluation with independently reviewed benchmarks.
Current / target	2 / 4
Evidence	Student report of stale or less relevant recall
Rationale	The problem was observed and explained. No specific Tesseract retrieval correction was implemented.
Next action	Build dated and contradictory memory cases with expected answers.

Competency rubrics - continued

A5 - Autonomous operation

Observable behavior	Schedule background work with clear behavior and visible failures.
1	Cannot yet run a background task reliably in a guided setup.
2	Runs a manual task but repeated execution needs intervention.
3	Demonstrates scheduled operation and explains its intended behavior.
4	Records repeated runs and recovers from a realistic interruption.
5	Designs unattended operation validated across independently maintained installations.
Current / target	3 / 4
Evidence	Student report of daily maintenance and fewer interruptions
Rationale	The personal setup is reported working. One-time setup for a wider user population is an untested usability hypothesis.
Next action	Record scheduled runs, exceptions, and a failure-recovery example.

T1 - Exploratory testing

Observable behavior	Challenge a system with varied inputs and record what happens.
1	Cannot yet suggest a meaningful variation in input.
2	Tries variations without recording their purpose or result.
3	Targets a failure area and explains an observed problem.
4	Systematically explores boundaries and preserves reproducible findings.
5	Leads risk-based exploratory sessions and improves others' test coverage.
Current / target	3 / 4
Evidence	Student Tesseract stress-testing account
Rationale	The student identified interruption and retrieval problems through varied personal use.
Next action	Turn one exploratory session into a dated issue log.

Competency rubrics - continued

T2 - Failure reproduction

Observable behavior	Capture conditions that reliably produce a reported defect.
1	Cannot yet describe a failure beyond saying it did not work.
2	Describes a failure but lacks sufficient steps to reproduce it.
3	Supplies repeatable steps and separates symptom from cause.
4	Minimizes intermittent failures and confirms cause with controlled changes.
5	Creates reusable diagnostic methods for complex system failures.
Current / target	2 / 4
Evidence	Student recall and approval-interruption reports
Rationale	Symptoms are clear, but no captured failing query or approval trace was supplied in this exchange.
Next action	Save minimal inputs, environment details, and expected versus actual output.

T3 - Automated regression

Observable behavior	Use repeatable checks to protect behavior after changes.
1	Cannot yet execute or interpret a guided automated check.
2	Runs a suite with help but cannot explain its assertions.
3	Adds or explains a regression case and interprets its result.
4	Maintains representative regression coverage across releases.
5	Designs evaluation infrastructure and audits misleading or weak assertions.
Current / target	Not assessed / 4
Evidence	E1 and E3 automated test artifacts
Rationale	Tests exist in repositories. Personal execution and interpretation remain unassessed, rather than scored as low ability.
Next action	Personally run a focused suite and explain a failing and passing case.

Competency rubrics - continued

T4 - Retrieval evaluation

Observable behavior	Compare returned evidence against a defined correct or acceptable answer.
1	Cannot yet define a relevant answer for a sample query.
2	Judges individual results informally without a stable evaluation set.
3	Runs labeled queries and reports relevance or freshness outcomes.
4	Compares versions with defined measures and analyzes failures.
5	Publishes reproducible evaluations with independent review and contamination controls.
Current / target	2 / 4
Evidence	Student report of stale or irrelevant memory
Rationale	Informal recall challenges were reported. No measured relevance or freshness score is established.
Next action	Create 20 queries covering current facts, changes, conflicts, and unknowns.

T5 - User and player testing

Observable behavior	Observe another person using a flow and revise from evidence.
1	Cannot yet define an observable task for a tester.
2	Collects general opinions without observing task behavior.
3	Observes a task, records difficulties, and makes a justified revision.
4	Repeats testing across user types and checks whether revisions help.
5	Leads a rigorous user-testing process with defensible outcomes.
Current / target	Not assessed / 3
Evidence	E4 next-step statement and E6 gameplay artifact
Rationale	Artifacts establish interfaces and a game. Fresh human test outcomes were not established in the reviewed material.
Next action	Observe two people performing a task and playing one short session.

5. Adaptive action plan

My strongest documented area is delivering and presenting projects. My vulnerable areas are measured retrieval quality, repeatable failure reproduction, and evidence of individual programming depth. I will prioritize the Rust memory project because it addresses a problem I actually encountered. I will preserve game development as a connected direction, using a small prototype rather than beginning a large game alongside the retrieval rebuild. The previously recorded planning limit is 3-4 project hours weekly and $20-$60 monthly total AI spending. I will reconfirm those limits before expanding scope. No new hardware purchase or subscription is assumed.

Weeks 1-2 after review - Define a memory evaluation set

20 labeled queries covering current facts, older facts, contradictions, irrelevant matches, and unknown answers. Record expected evidence and output before changing retrieval.

Constraint: 2 weekly sessions within the previously recorded 3-4 project hours. Axes: M5, A4, T2, T4.

Weeks 3-4 - Test the current baseline

Run the same queries, record source dates and relevance judgments, and write a reproducible account of at least one stale-result failure. Record maintenance runs and interruption causes separately.

Constraint: No new paid service assumed. Axes: A2, A5, T1, T4.

Weeks 5-8 - Build a bounded Rust retrieval prototype

Use a small synthetic dataset. Store event time, record time, source, and update links. Test verified updates, incorrect newer statements, and unresolved conflicts against the baseline.

Constraint: Keep full Tesseract feature parity outside this milestone. Axes: M1, M2, A4, T3.

Weeks 9-12 - Validate and communicate

Ask another person to run setup or recall tasks. Document results and my contribution. Build one small game or creative interaction only if the retrieval milestone is complete and time permits.

Constraint: Retain school and family priorities. Axes: M3, M7, M8, T5.

Each milestone through May 2028 - Review the chart

Update scores only when the rubric and evidence support a change. Recheck relevant job expectations, revise the portfolio, and select the next gap.

Constraint: Reconfirm available time and budget before expansion. Axes: All axes.

Review scores after each milestone. Keep recording dates separate from event dates. Retain uncertainty when a newer statement cannot be verified.

6. Annotated research

S1 - The Future of Jobs Report 2025 - Skills outlook

World Economic Forum. 2025. Core reading. Accessed November 11, 2026.

The employer survey anticipates changing skill needs through 2030 and emphasizes analytical, technical, creative, and adaptive capabilities. I use it to justify a mixed competency map, while treating employer expectations as forecasts rather than guaranteed outcomes.

Open source

S2 - Anthropic Economic Index: AI's impact on software development

Anthropic. 2025-04-28. New independent research. Accessed November 11, 2026.

This analysis examines 500,000 coding interactions and distinguishes task automation from collaboration with developers. It helps me examine how implementation work could change, but it does not measure final code quality or prove productivity gains.

Open source

S3 - GDC 2026 State of the Game Industry Reveals Impact of Layoffs, Generative AI, and More

Beth Elderkin. 2026. New independent research. Accessed November 11, 2026.

The official survey summary describes AI use, employment pressures, and contrasting attitudes among game-industry professionals. I use it to challenge the assumption that easier production automatically means easier entry into games, while recognizing that respondents do not represent every studio.

Open source

S4 - Applied AI Demo Engineer

Anthropic. n.d.. New role-specific source. Accessed November 11, 2026.

This employer listing combines LLM applications, retrieval, integrations, reliable prototypes, and technical communication. I use its duties as one concrete role comparison rather than an entry-level promise or a universal definition of AI engineering.

Open source

Shared Zotero saving is pending authentication. The RIS import aid contains the four records and annotations. No Zotero item links are claimed.

7. Project evidence register

E1 - Tesseract MCP contribution | Merged contribution

Concurrent note creation and configurable agent identity. PR #21 by Rayhanson-Git merged November 4, 2026. Source and tests inspected. No new test run in this review.

Open project evidence

E2 - Tesseract macOS/Linux proposal | Contribution proposed

Portability, scheduling, and workflow guidance. PR #10 itself is not marked merged. Similar upstream changes must not be attributed solely from this PR.

Open project evidence

E3 - JimBro | Published

Integrated scheduling, workout logging, local storage, backups, and animated character. Core source and live entry screen inspected. November 10 QA record reports 49 domain checks and deployed Chromium journeys. Those checks were not rerun in this review.

Open project evidence

E4 - GymLog | Published prototype

Post-set logging, editable values, confirmation, and correction paths. Sample data lasts for the page session. Portfolio describes participant testing as a next validation step.

Open project evidence

E5 - LeaveBy | Published

Backward scheduling with input validation and date calculations. Source and live form inspected. Uses user estimates rather than live transit information.

Open project evidence

E6 - Wasp Dodge+ | Published browser game

Movement, collision detection, four lives, cooldown, scoring, and restart logic. Source and live game menu inspected. Full playthrough and player testing were not performed in this review.

Open project evidence

E7 - Gesture Particles | Published

MediaPipe landmarks mapped to Three.js particle behavior. Source and live interface inspected. Camera-based behavior was not tested in this review. Integrates an existing model.

Open project evidence

E8 - Virtual Theremin | Published

Hand tracking mapped to pitch, volume, bass, and drum triggers. MediaPipe and Tone.js integration inspected. Live start screen opened. Camera and audio performance were not tested.

Open project evidence

E9 - Neon Orbit Arcade | Published

Particles, portals, forces, and Calm/Chaos modes. Source and rendered live canvas inspected.

Open project evidence

E10 - Orbit Playground | Published

Orbiters, changing palettes, adjustable speed, and input response. Source and rendered live canvas inspected.

Open project evidence

E11 - Studio Apartment Evening | Published audio

Audio storytelling and multitrack editing in Adobe Audition. Published 186-second audio loaded and session screenshot inspected. The page credits 11 self-recorded sounds. No listening assessment was performed in this review.

Open project evidence

E12 - Portfolio website | Published

Case studies, visual identity, responsive navigation, and project presentation. Nine case studies and core portfolio interaction code inspected.

Open project evidence

E13 - Devil May Cry music video | Completed, student-reported

Video editing and game-related storytelling. Not yet published on the portfolio. Media and editing timeline not inspected.

E14 - Rust retrieval system | Planned

Time-aware memory, conflict resolution, and retrieval evaluation. No working Rust implementation established in this conversation.

Before submission

Confirm proposed scores and targets, complete Zotero saving, and verify the current Brightspace instructions. Earlier 1c scores remain historical and use different axes. The separate course collaboration record documents the preparation process.